The ALLEGRO trial:

A placebo controlled r**A**ndomised trial of intravenous **L**idocaine in acce**LE**rating **G**astrointestinal **R**ecovery after c**O**lorectal surgery

Statistical Analysis Plan

- Minimally Invasive RCT -

A UK Collaborative Trial funded by NIHR HTA

Version: 2

Version Date: 9th September 2024

Based on
Protocol Version: Version 8. Go to ALLEGRO (abdn.ac.uk). scroll to Key Facts about the Study and document (SHOW MORE) and download ALLEGRO_Protocol (V8.0 21st December 2022)

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Signatures

By signing this document, I am confirming that I have read, understood and approve the statistical analysis plan (SAP) for the **ALLEGRO** trial.

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Version History

SAP version	Protocol version	Section number changed	Description of and reason for change	Date changed
Version 1	Version 7		New document	<<date>>
Version 2	Version 8	11.3	Primary outcome was analysed using log link. There was an error in section 11.3 where it is stated that logit link will be used. This change has been made after the analysis was completed.	09/09/2024
Version 2	Version 8	11.5	ERAS compliance subgroup was pre-planned and it was an error of omission from SAP. It has been included in the SAP after looking at the data. This change has been made after the analysis was completed.	09/09/2024

Glossary of Abbreviations

ACCORD	Academic and Clinical Central Office for Research & Development - Joint office for The University of Edinburgh and Lothian Health Board
AE	Adverse Event
AR	Adverse Reaction
AUC	Area Under the Curve
CHaRT	Centre for Healthcare Randomised Trials
CI	Confidence Interval
CONSORT	Consolidated Standards of Reporting Trials
CRF	Case Report Form
CRP	C-reactive Protein
CSR	Clinical Study Report
DMC	Data Monitoring Committee
EQ-5D	EuroQol Group's 5-dimension health status questionnaire
ERAS	Enhanced Recovery After Surgery
GI-2/3	Gastrointestinal Recovery endpoints (see protocol for definitions)
HSRU	Health Services Research Unit
HTA	Health Technology Assessment
NIHR	National Institute of Health Research
ITT	Intention-to-Treat
POD	Post Operative Day
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SD	Standard Deviation

Table of Contents

ALLEGRO

This is the statistical analysis plan (SAP) and accompanying documents detailing the statistical analyses planned for the ALLEGRO Trial. The SAP is based on the protocol version 8 and any deviations from the plan will be described.

1 Introduction

Delayed return of gut function is the most common cause of delayed discharge from hospital after elective colorectal surgery. Although self-limiting it prolongs hospital stay by a few days in most cases but can last up to 10 days if severe. Affected patients cannot tolerate oral intake due to nausea and vomiting requiring intravenous fluids and ongoing opioid analgesia. In comorbid, frail or elderly patients the development of gut dysfunction can contribute to other major complications including pulmonary aspiration, pneumonia and acute kidney impairment.

There are no specific preventative or therapeutic interventions to induce return of gut function in current use. From the evidence available, intravenous lidocaine may improve return of gut function and is the motivation for this placebo controlled r**A**ndomised trial of intravenous **L**idocaine in acce**LE**rating **G**astrointestinal **R**ecovery after c**O**lorectal surgery (ALLEGRO). ALLEGRO comprises two RCTs – an RCT within minimally invasive surgery and an RCT within open surgery.

This SAP covers the minimally invasive RCT only. The open RCT will be addressed in a separate analysis plan.

2 Study Aims and Objectives

The **aim** of this trial is to measure whether perioperative intravenous lidocaine achieves faster return of gut function for more patients after colorectal surgery.

The ***primary objective*** is to compare the proportion of randomised subjects between IV lidocaine and placebo that have achieved return of gut function at 72 hours postoperatively.

The ***secondary objectives*** are to measure whether perioperative intravenous lidocaine is effective in achieving faster postoperative return of gut function along with other recovery metrics including use of analgesia, length of stay, quality of recovery, vomiting, complications, quality of life and safety.

3 General Study Design

More detail is given in the trial protocol.

- *Type of design* – A U.K. multicentre double blind superiority parallel group placebo-controlled randomised CTIMP of a 6 or 12-hour perioperative infusion of intravenous lidocaine in colorectal surgery.
- The primary study will be in minimally invasive (laparoscopic/robotic) colectomy and the sample size of 562 is calculated accordingly (see later).
- *Study hypothesis* is to assess if perioperative IV lidocaine will reduce delayed return of gut function after elective colorectal surgery.

4 Interventions to be evaluated

- The active intervention is a 6 or 12-hour (both permissible) perioperative infusion of intravenous lidocaine– or Investigational Medicinal Product (IMP).
- The placebo arm will receive 0.9% sterile Sodium Chloride solution for injection (a clear and colourless solution appearing to be identical to the IMP) at the equivalent time point and for equivalent duration.

5 Change of Status

Full inclusion and exclusion criteria are described in the Protocol (section 4.2 and 4.3) along with change of status definitions (section 5.6).

Any participant randomised that subsequently becomes ineligible before surgery started and before IMP is given will be classified as a post-randomisation exclusion. We will record and report these post-randomisation exclusions, they will not contribute to the sample size and we aim to replace them by subsequent randomisations.

Participants who decline further follow-up will be classified as such, their subsequent data will be missing, and they will not be replaced by over recruitment.

6 Randomisation, Allocation and Blinding

All participants who agree to enter the study will be logged with the central trial office and given a unique Study Number. Randomisation will utilise the existing proven remote automated computer randomisation application in the central trial office in the Centre for Healthcare Randomised Trials (CHaRT, a fully registered UK CRN clinical trials unit) in the Health Services Research Unit (HSRU), University of Aberdeen.

The unit of randomisation will be the participant and on a 1:1 ratio. Randomisation will be computer-allocated and minimised by:

- Age (<50 years, 50-74 years, 75 years and older)
- Gender
- Trial centre

Participants, medical staff and study staff/outcome assessors will all be blinded in this study, but the statistician will not be blinded to treatment allocation. All participants will be anonymised in the analysis database.

7 Outcome Measures

7.1 Primary Outcome(s)

The primary outcome is achievement (yes/no) of GI-3 recovery. This is a binary composite endpoint defined as achievement of both of the following two events: tolerating diet (defined as ingestion of food and drink without significant nausea or vomiting for 3 consecutive meals) and passage of flatus OR stool (whichever comes first) at 72 hours after the start of operation.

7.2 Secondary Outcomes

- Time to return of gut function using the GI-3 recovery definition (a composite endpoint defined as time from surgery to the later time to establish both of the following two events: tolerating diet without significant nausea or vomiting for 3 consecutive meals AND first passage of flatus or stool).
- Time to return of gut function using the GI-2 recovery definition (a composite endpoint defined as time from surgery to the later time to establish both of the following two events: tolerating diet without significant nausea or vomiting for 3 consecutive meals AND first passage of stool)

- Prolonged Postoperative Ileus (PPOI= failure to establish GI-3 by 120 hours after surgery (postoperative day 5))
- Nausea and vomiting
- Quality of analgesia
- Total postoperative opioid consumption
- Quality of Recovery
- Quality of life assessment
- Measurement of specific enhanced recovery guideline variables that have been shown to impact GI recovery
- Time to achievement of medical criteria for discharge
- Patient-reported assessment of readiness for discharge
- Health economic evaluation

7.3 Tertiary Outcomes*

In exception to total length of stay, these outcomes are mainly for exploratory/safety analyses and although reported and summarised have no statistical testing performed.

- Total length of stay
- 30- and 90-day mortality
- Unplanned re-admissions within 30 days of date of operation
- Major complications (defined by Clavien-Dindo classification grade 3+)

*In the published protocol, these are listed as secondary outcomes.

8 Schedule of data collection

Assessment	Recruitment /Baseline	POD 1	POD 2	POD 3	Daily from POD 4 until discharge	POD 7	Discharge	30 days	90 days
Assessment of Eligibility Criteria	✓								
Written informed consent	✓								
Pregnancy Test (where applicable)	✓								
Demographic data, contact details	✓								
Clinical history/ past medical history	✓								
Drug history esp. laxatives	✓								
Height	✓								
Weight	✓	✓							
p-POSSUM[D]	✓	✓ (part)						✓ (part)	
Operation type		✓							
Duration of operation		✓							
Blood loss (mls)		✓							
OBAS score[D]	✓	✓	✓	✓	✓	✓			
QoR score[D]	✓	✓	✓	✓	✓	✓		✓	
PONV score[D]		✓	✓	✓	✓				
EQ5D[D]	✓	✓	✓	✓	✓	✓		✓	✓
CRP	✓		✓	✓					
Time to first flatus			✓	✓	✓				
Time to first bowel movement			✓	✓	✓				
Time to tolerating solid food			✓	✓	✓				
antiemetic dose total up to 72 hours							✓		
Total number of episodes vomiting		✓	✓	✓					
Total opioid consumption in-hospital up to 72 hours							✓		
ERAS protocol compliance		✓	✓	✓					

Assessment	Recruitment /Baseline	POD 1	POD 2	POD 3	Daily from POD 4 until discharge	POD 7	Discharge	30 days	90 days
Achievement of medical criteria for discharge Y/N				✓	✓				
Patient-reported readiness for discharge				✓	✓				
Length of stay (days)							✓		
Complications							✓	✓	
Unplanned readmissions								✓	✓
Use of primary care services								✓	
Mortality							✓	✓	✓
Adverse events		✓	✓	✓	✓	✓		✓	

POD: post operative day; [D] derived scores with validated code

9 Adverse Events

Each initial adverse event (AE) will be considered for severity, causality or expectedness. A serious adverse event (SAE) is any AE that:

- Results in death;
- Is life threatening;
- Required hospitalisation or prolongation of existing hospitalisation;
- Results in persistent or significant disability or incapacity;
- consists of a congenital anomaly or birth defect;
- results in any other significant medical event not meeting the criteria above.

Please see the study protocol for more details on AEs. The number of adverse events (AEs) and serious adverse events (SAEs) and the proportion of participants with an event will be presented. These will be tabulated but not formally analysed and will be summarised by Intention-to-Treat (ITT) and as treated. Safety data will be presented for all participants who were randomised and received the IMP.

10 Sample Size and Power Calculation

A sample size of 562 randomised 1:1 to IV lidocaine or placebo will have 90% power at a 2-sided 5% level of significance to detect a relative reduction of 33% from 40% to 26.8% (absolute reduction of 13.2%) in non-return of gut function at 3 days post-op. If the event rate is lower, the same power should detect a 40% relative reduction from 30% to 18% (a 12% absolute reduction). The sample size base based on the assumption that there should be no missing primary outcome data; however, in reality, some participants withdraw from having outcome data collected before reaching the primary outcome timepoint, so there may be some missing primary outcome data.

11 Statistical Methods

11.1 General Methods

All the analyses will be based on the Intention-to-Treat (ITT) principle. Final analysis will take place after full recruitment and follow-up. The results of the trial will be presented following the standard CONSORT recommendations[1]. Baseline and follow-up data will be summarised using the appropriate descriptive statistics and graphical summaries. Treatment effects will be presented with 95% confidence intervals (CIs) (apart from subgroup analysis). There will be no adjustment to secondary outcomes for multiple testing. All eligible participants who provided consent will be included in the analysis. Post-randomisation exclusions will be reported as such and not included in any analysis. We aim to replace these post-randomisation exclusions if it is possible within the recruitment window.

Full details of analyses for each outcome variable along with sample Stata code is available in appendix.

11.2 Interim Analysis

No interim analysis is planned for the Minimally Invasive study. However, a regular report is produced for the independent Data Monitoring Committee (DMC) which monitors trial progress and specifically any safety issues.

11.3 Primary Outcome

The primary outcome will be analysed using a generalised linear model with a log link function, given its binary state for each individual. Treatment effects will be reported for the primary endpoint only (by 72 hours). The generalised linear model will be used to adjust for the minimisation variables; gender, age and centre (the latter via a random effect).

11.4 Secondary Outcomes

Secondary outcomes will be analysed in a similar manner but using the appropriate statistical model for the outcome (Cox regression for time-to-event outcomes, linear regression for continuous variables) with adjustment for minimisation variables; gender, age and centre (the latter via a random effect). OBAS, QoR-15 and EQ-5D will be analysed using repeated measures including baseline score as an additional covariate. Treatment effects will be estimated from time-by-treatment interactions at each time point.

11.5 Subgroup Analyses

Subgroup by treatment interaction will be assessed for the primary outcome by including interaction terms in the models outlined above. These have been assessed a-priori to be:

- 6 hour vs. 12 hour IMP infusion (defined by what was intended)
- By operation (right v non-right hemicolectomy)
- Gender (M vs F)

- Age (<50 years, 50-74 years, 75 years and older)
- ERAS protocol compliance (High, moderate, low)

A stricter level of statistical significance (2-sided 1% significance level) will be applied to these analyses given their exploratory nature. Corresponding 99% confidence intervals will therefore be calculated.

11.6 Sensitivity analysis

As per recommendation[2], a sensitivity analyses on the primary outcome will include multiple imputation such as MICE and/or we will explore a range of values for missing data imputed under missing not at random assumptions (such as pattern mixture models) if the missingness is >10%. The extent of missingness will be assessed along with a determination of if the data are MAR or MCAR.

As Edinburgh is the sponsor site, it is expected to have the largest recruitment. Therefore, a sensitivity analysis comparing Edinburgh with the rest of the centres will be performed.

11.7 Compliance

There are two levels of compliance within ALLEGRO: (i) patients who were eligible to receive IMP/placebo, but who did not and (ii) patients who did not receive the intended duration of IMP/placebo. Both will be explored in a compliance analysis.

11.8 Missing Data

11.8.1 Missing Outcome Data

The sensitivities of treatment effect estimate to missing all primary outcome data will be explored; these models will explore the robustness of the treatment estimate to whatever small amount of missing data there is. We will follow the strategy outlined in White *et al*[2]. The analysis will use all available data that we believe are valid under the assumption of missing at random. We will then use a suite of sensitivity analysis to explore the robustness of the primary outcome to departures from assumption, including all randomised participants. If required, sensitivity analyses will include multiple imputation, and imputing a range of values for missing data under missing not at random assumptions.

11.8.2 Missing Baseline Data

Data missing at baseline will be reported as such. However, if baseline is required as a covariate in a treatment-time model but is missing this will be imputed based on the overall mean of that variable.

11.9 Statistical software

All analysis will be carried out in Stata 17[4] or higher.

11.10 Derived variables – – Composite outcome measures and patient reported outcome measures (PROMS)

GI-2 and GI-3 recovery are composite endpoints. Table 16.2 shows how each score will be calculated.

PROMS are several patients-reported outcomes collected using validated questionnaires which require scores and/ or subscales to be calculated. These are indicated by ^{D} in table above in the Timing of Outcome Measures (Section 8). Codes developed in-house are checked and validated by an independent statistician using dummy data. Missingness for amalgamated scores will be treated according to decisions made by the Project Team on informed by a review of how others have treated missingness for these derived variables. Table 16.1 shows how each score will be calculated.

12 COVID-19

The effect of COVD-19 will be explored. In the first instance, periods before, during and after COVID-19 will be summarised using appropriate descriptive statistics and graphical summaries. If need be, formal analysis will be carried out to explore the effect of COVID-19.

13 Dummy Tables

Table 13.1 Overall Recruitment and by centre

Centre	Patients randomised N=	Randomised to IV lidocaine N=	Randomised to placebo N=
Overall recruitment			
Recruitment by centre			
Addenbrooke's hospital			
...			

Values are n(%)

Table 13.2 Reasons why participants were not randomised

	N=
Reasons for ineligibility	
Planned epidural anaesthesia	
Planned regional or local continuous infusion of lidocaine at the same time as lidocaine infusion	
Planned open surgery	
……..	
Reasons for declining to take part	
Did not want to be randomised	
….	

Values are n

Table 13.3 Baseline minimisation variables

	IV lidocaine N=	Placebo N=
Centre		
…….		
……..		
Age		
< 50		
50 -74		
≥ 75		
Gender		
Male		
Female		

Values are n(%)

Table 13.4 Baseline characteristics

	IV lidocaine N=	Placebo N=
Age		
n: mean(sd)		
median(IQR)		
(min, max)		
Current smoking status, n(%)		
Yes		
No		
Missing		
BMI		
n: mean(sd)		
median(IQR)		
(min, max)		
Possum Score		
Physiological score (12-88)		
n: mean(sd)		
median(IQR)		
(min, max)		
Operative severity score (6-48)		
n: mean(sd)		
median(IQR)		
(min, max)		
Mortality prediction %		
n: mean(sd)		
median(IQR)		
(min, max)		

Table 13.5 Surgical procedures

	IV lidocaine N=	Placebo N=
Operation Performed, n(%)		
Right hemicolectomy		
Extended Right hemicolectomy		
Left hemicolectomy		
Sigmoid colectomy		
Subtotal colectomy		
(ileo-sigmoid/ileo-rectal anastamosis)		
High anterior resection		
Other		
Unplanned Stoma, n(%)		
Yes		
No		
Missing		
Surgical approach used, n(%)		
Laparoscopic/robotic		
Conversion from laparoscopic/robotic to open		
Missing		
Robotic surgery used, n(%)		
Intraoperative blood loss recorded (mls)		
N: mean (SD)		

median(IQR)		
(min, max)		

Table 13.6 Baseline OBAS, quality of recovery-15 and quality of life (EQ-5D-5L)

Variable	**IV lidocaine N=**	**Placebo N=**
OBAS domains		
Pain		
n: mean(sd)		
median(IQR)		
(min, max)		
Vomiting		
n: mean(sd)		
median(IQR)		
(min, max)		
Itching		
n: mean(sd)		
median(IQR)		
(min, max)		
Sweating		
n: mean(sd)		
median(IQR)		
(min, max)		
Freezing		
n: mean(sd)		
median(IQR)		
(min, max)		
Dizziness		
n: mean(sd)		
median(IQR)		
(min, max)		
Satisfaction with pain management		
n: mean(sd)		
median(IQR)		
(min, max)		
Total		
n: mean(sd)		
median(IQR)		
(min, max)		
Quality of recovery-15		
n: mean(sd)		
median(IQR)		
(min, max)		
EQ-5D-5L		
Total score		
n: mean(sd)		
median(IQR)		
(min, max)		
Visual analogue scale		
n: mean(sd)		
median(IQR)		
(min, max)		

Table 13.7 Proportion of participants who have achieved return of gut function.

	IV lidocaine N=	Placebo N=
Achieved GI-3 Status by 72 hours post-operative		
Achieved GI-2 Status by 96 hours post-operative		

Values are n(%)

Table 13.8 Analysis of the primary outcome of proportion of participants who have achieved return of gut function at 72 hours post-operatively

Analysis	Estimate* (95%CI), p-value
Intention to treat	
Unadjusted	
Adjusted for minimisation variables	
Per-protocol	
Unadjusted	
Adjusted for minimisation variables	

*Generalised linear model with a log link function

Table 13.9 Time to return of gut function using the GI-2 and GI-3 recovery function

Analysis	Estimate*(95%CI), p-value
GI-2	
Unadjusted	
Adjusted for minimisation variables	
GI-3	
Unadjusted	
Adjusted for minimisation variables	

*Cox regression model. # If proportional hazard assumption is violated, restricted mean survival time will be used.

Table 13.10 Prolonged post-operative Ileus – proportion of patients who failed to establish GI-3 by 120 hours after surgery.

Variable	IV lidocaine N=	Placebo N=	Model	Estimate*(95%CI), p-value
Prolonged post-operative Ileus, n(%)			Unadjusted Adjusted#	

*Generalised linear model with a log link function, # Adjusted for minimisation variables

Table 13.11 PONV impact scale score by treatment arms

Day	IV lidocaine N=		Placebo N=	
	<5	≥5	<5	≥5
1				
2				
3				
…				

Values are n(%), *≥5 defines clinically important PONV

Table 13.12 Descriptive of episodes of vomiting by treatment arms

	IV lidocaine N=	Placebo N=
Day 1		
n: mean(sd)		
median(IQR)		
(min, max)		
Day 2		
n: mean(sd)		
median(IQR)		
(min, max)		
Day 3		
n: mean(sd)		
median(IQR)		
(min, max)		
Day 4		
n: mean(sd)		
median(IQR)		
(min, max)		
Total cumulative episodes in 72 hours		
n: mean(sd)		
median(IQR)		
(min, max)		

Table 13.13 Descriptive of rescue anti-emetics used

	IV lidocaine N=	Placebo N=
Intraoperative, dose (mgs)		
Ondansetron		
n: mean(sd)		
median(IQR)		
(min, max)		
Cyclizine		
n: mean(sd)		
median(IQR)		
(min, max)		
Prochlorperazine		
n: mean(sd)		
median(IQR)		
(min, max)		
Postoperative, dose (mgs)*		
Ondansetron		
n: mean(sd)		
median(IQR)		
(min, max)		
Cyclizine		
n: mean(sd)		
median(IQR)		
(min, max)		
Prochlorperazine		
n: mean(sd)		
median(IQR)		
(min, max)		
Total anti-emetics, dose (mgs)		
Ondansetron		
n: mean(sd)		
median(IQR)		
(min, max)		
Cyclizine		
n: mean(sd)		
median(IQR)		
(min, max)		
Prochlorperazine		
n: mean(sd)		
median(IQR)		
(min, max)		

*This will be calculated using total (Sec K, Q3) minus intra-operative (Sec C, Q10)

Table 13.14 Overall benefit of analgesic score (OBAS) ratings

Time since surgery	OBAS domains						
	Pain	**Vomiting**	**Itching**	**Sweating**	**Freezing**	**Dizziness**	**Satisfaction with pain management**
Postoperative Day 1							
IV lidocaine (n=)							
Placebo (n=)							
Postoperative Day 2							
IV lidocaine (n=)							
Placebo (n=)							
Postoperative Day 3							
IV lidocaine (n=)							
Placebo (n=)							
Postoperative Day 4							
IV lidocaine (n=)							
Placebo (n=)							
Postoperative Day 5							
IV lidocaine (n=)							
Placebo (n=)							
Postoperative Day 6							
IV lidocaine (n=)							
Placebo (n=)							
Postoperative Day 7							
IV lidocaine (n=)							
Placebo (n=)							

Data presented as mean ± SD. *The five-point scales for the domains are as follows: Pain: 0 = minimal pain, 4 = maximum imaginable pain; Vomiting, Itching, Sweating, Freezing and Dizziness (regarding distress and bother from symptoms): 0 = not at all, 4 = very much; Satisfaction with pain management: 0 = not at all, 4 = very much

Table 13.15 Total OBAS score analysis

	IV lidocaine N=	Placebo N=	Estimate (95% CI); p-value
Baseline			
Postoperative Day 1			
Postoperative Day 2			
Postoperative Day 3			
Postoperative Day 4			
Postoperative Day 5			
Postoperative Day 6			
Postoperative Day 7			

Data presented as mean ± SD. Linear mixed model will be used to analyse the repeated measures of OBAS outcome. Treatment effects at each time were derived from the interaction term for time by treatment.

Table 13.16 Descriptive of post-operative opioid consumption

		IV lidocaine N=	Placebo N=
Intraoperative analgesia			
NSAIDS			
Morphine (mgs)			
Fentanyl (mcgs)			
Alfentanil (mcgs)			
Remifentanil (mcgs)			
Oxycodone (mgs)			
Diamorphine (mgs)			
Tramadol (mgs)			
Postoperative analgesia			
NSAID			
Morphine parenteral	PCA (mgs)		
	Subcutaneous (mgs)		
Morphine Oral	Modified release (MR) (mgs)		
	Immediate release (IR) (mgs)		
Fentanyl PCA (mgs)			
Fentanyl patch (mgs)			
Oxycodone parenteral	PCA (mgs)		
	Subcutaneous (mgs)		
Oxycodone Oral	Modified release (MR) (mgs)		
	Immediate release (IR) (mgs)		
Tramadol Oral	Modified release (MR) (mgs)		
	Immediate release (IR) (mgs)		
Codeine (mgs)			
Dihydrocodeine(mgs)			

Data presented as mean ± SD

Table 13.17 Quality of recovery (QoR-15) score analysis

	IV lidocaine N=	Placebo N=	Estimate (95% CI); p-value
Baseline			
Postoperative Day 1			
Postoperative Day 2			
Postoperative Day 3			
Postoperative Day 4			
Postoperative Day 5			
Postoperative Day 6			
Postoperative Day 7			
Postoperative Day 30			

Data presented as mean ± SD. Linear mixed model will be used to analyse the repeated measures of QoR-15 outcome. Treatment effects at each time were derived from the interaction term for time by treatment.

Table 13.18 Quality of life (EQ-5D) assessment analysis

	IV lidocaine N=	Placebo N=	Estimate (95% CI); p-value
Baseline			
Postoperative Day 1			
Postoperative Day 2			
Postoperative Day 3			
Postoperative Day 4			
Postoperative Day 5			
Postoperative Day 6			
Postoperative Day 7			
Postoperative Day 30			
Postoperative Day 90			

Data presented as mean ± SD. Linear mixed model will be used to analyse the repeated measures of EQ-5D outcome. Treatment effects at each time were derived from the interaction term for time by treatment.

Table 13.19 Quality of life (EQ-5D) assessment - Visual analogue scale

	IV lidocaine N=	Placebo N=	Total N=
Baseline			
Postoperative Day 1			
Postoperative Day 2			
Postoperative Day 3			
Postoperative Day 4			
Postoperative Day 5			
Postoperative Day 6			
Postoperative Day 7			
Postoperative Day 30			
Postoperative Day 90			

Data presented as mean ± SD.

Table 13.20 Descriptive of measurement of specific enhanced recovery guideline (ERAS)

	IV lidocaine N=	Placebo N=
IV Dexamethasone given at anaesthesia induction		
Use of intrathecal (spinal) diamorphine or other opiate		
PONV prophylaxis prescribed regularly for first 48 hours		
Laxative prescribed postoperatively		
Nasogastric tube placed intraoperatively and still in situ when patient woke up		
Chewing gum prescribed postoperatively		
Total intravenous fluids, mean (SD)		
0.18% NaCl & 4% Dextrose		
Plasmalyte		
Ringer's Lactate		
Hartmann's		
NaCl 0.9%		
Dextrose 5%		
Colloid		
Packed red cells		
Other		
Preoperative carbohydrate loading given on day of surgery		
Mobilisation target achieved on Day 1[£]		
Patient offered food on Day 1		
Received post-op supplement drinks on day of surgery		
Patient weight on post-op Day 1, mean (SD)		
IV fluids discontinued within 48 hours of start of operation		
Urinary catheter removed within 48 hours of start of operation		
Mobilisation target achieved on Day 2[^]		

*Numbers in cells are n(%) except where indicated.
[£]Mobilisation target – 2 hours out of bed on day of surgery and 4-6 hours on day 1
[^]Mobilisation target – out of bed for at least 4 hours on day 2

Table 13.21 Time to achievement of medical criteria for discharge[£]

Analysis	Estimate*(95%CI), p-value
Unadjusted	
Adjusted for minimisation variables	

*Generalised linear model [£]Patient should meet all 5 criteria: independent hydration/nutrition, adequate analgesia by oral route, independent mobilisation, return of gut-function by GI-3 definition, no medical contraindication.

Table 13.22 Time to patient-reported readiness for discharge[£]

Analysis	Estimate*(95%CI), p-value
Unadjusted	
Adjusted for minimisation variables	

*Generalised linear model [£]Patient should meet all 6 criteria: independent hydration/nutrition, adequate analgesia by oral route, independent mobilisation, return of gut-function by GI-3 definition, no medical contraindication and patient happy to go home.

Table 13.23 30- and 90 days mortality and unplanned re-admissions within 30 days of date of operation

Outcomes	IV lidocaine N=	Placebo N=	Total N=
30 days mortality, n(%)			
90 days mortality, n(%)			
No of patients who had unplanned re-admission, n(%)			

*mean (SD) / median (IQR) – depending on the distribution

Table 13.24 Length of hospital stay within 30 days after surgery

Variable	IV lidocaine N=	Placebo N=	Model	Estimate*(95%CI), p-value
Total length of hospital stay, days$			Unadjusted	
			Adjusted#	

*Generalised linear model, $mean (SD) / median (IQR) – depending on the distribution, # Adjusted for minimisation variables

Table 13.25 Major complications

	IV lidocaine N=	Placebo N=	Total N=
Clavien-Dindo			
Number of patients who had complication of Clavien-Dindo grade 3 and above			
Number of events			
Respiratory complications			
Clavien IIIa			
Clavien IIIb			
Clavien IVa			
Clavien IVb			
Clavien V			
Infectious complications			
Clavien IIIa			
Clavien IIIb			
Clavien IVa			
Clavien IVb			
Clavien V			
Cardiovascular complications			
Clavien IIIa			
Clavien IIIb			

Clavien IVa			
Clavien IVb			
Clavien V			
Renal, hepatic, pancreatic and gastrointestinal complications			
Clavien IIIa			
Clavien IIIb			
Clavien IVa			
Clavien IVb			
Clavien V			
Surgical complications			
Clavien IIIa			
Clavien IIIb			
Clavien IVa			
Clavien IVb			
Clavien V			

Table 13.26 Summary of all adverse events – descriptors

		IV lidocaine N=	**Placebo N=**	**Total N=**
Number of participants with adverse events				
Number of participants without adverse events				
Severity				
	No Information			
	Mild			
	Moderate			
	Severe			
Relationship to Trial Drug				
	No Information			
	None			
	Possible			
Expectedness				
	No Information			
	Expected			
	Unexpected			
Outcome				
	No information			
	Resolved			
	Ongoing			

Table 13.27 Summary of all serious adverse events – descriptors

	IV lidocaine N=	Placebo N=	Total N=
Number of participants with SAEs			
Number of participants without SAEs			
Description of SAEs			
Congenital anomaly/ birth defect			
Required hospitalisation			
Resulted in persistent or significant disability			
Life-threatening			
Death			
Severity			
No Information			
Mild			
Moderate			
Severe			
Expectedness			
No Information			
Expected			
Unexpected			
Outcome			
Completely recovered			
Recovered with sequelae			
Death			

14 Dummy Figures

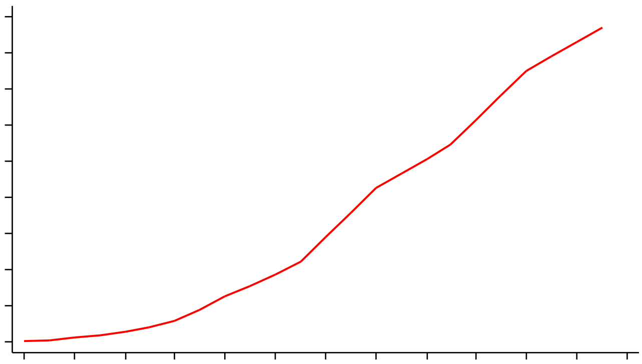

Figure 1 Recruitment graph

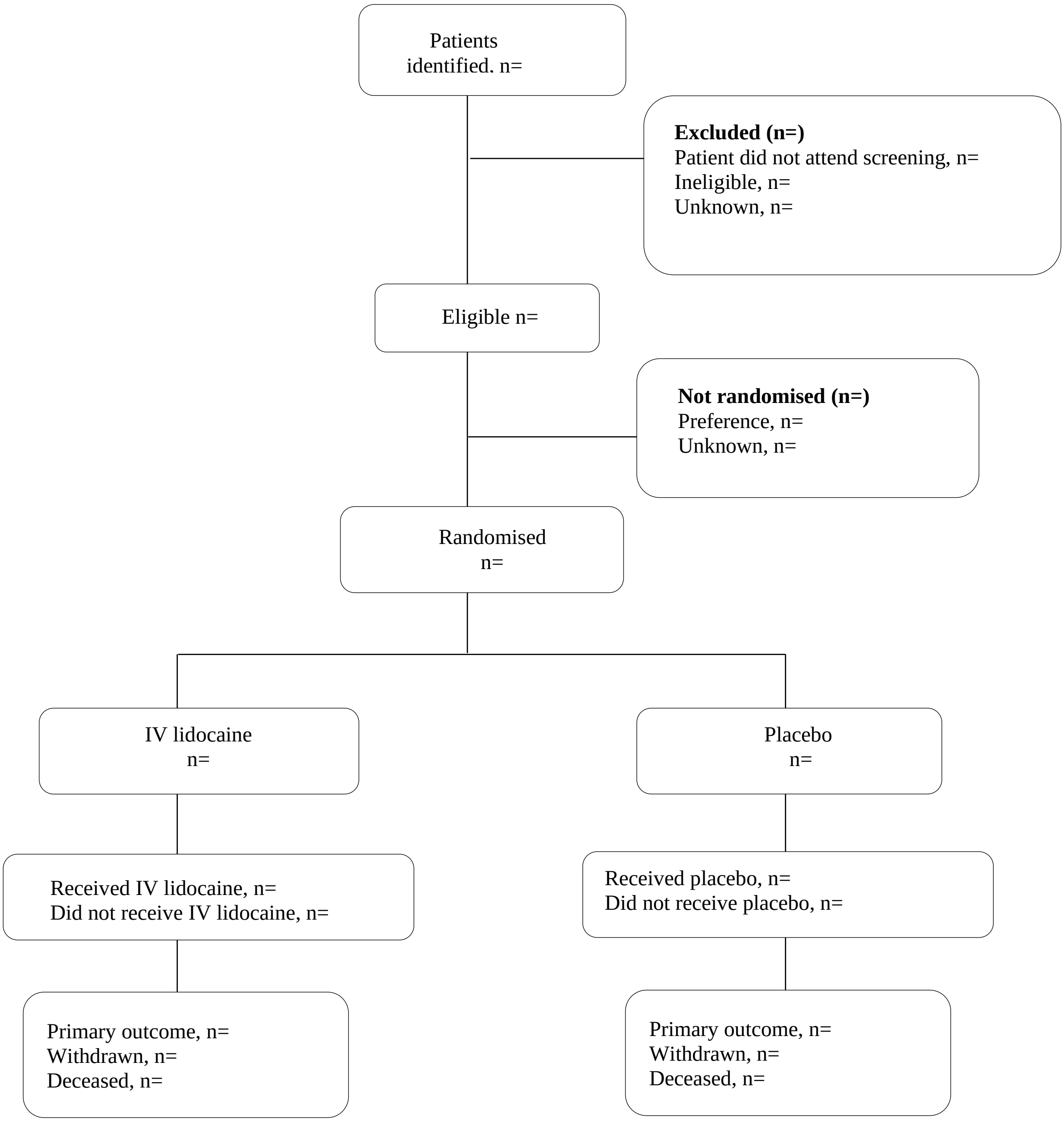

Figure 2 Participant flow

Figure 3 Subgroup analysis

15 References

1. Schulz KF, Altman DG, Moher D. CONSORT 2010 statement: Updated guidelines for reporting parallel group randomised trials. *BMJ*. 2010;340:c332. http://www.bmj.com/content/340/bmj.c332.abstract. doi: 10.1136/bmj.c332.

2. White IR, Horton NJ, Carpenter J, Pocock SJ. Strategy for intention to treat analysis in randomised trials with missing outcome data. *BMJ* 2011; 342:d40.

3. Thomas R Sullivan, Ian R White, Amy B Salter et al. Should multiple imputation be the method of choice for handling missing data in randomized trials? Statistical Methods in Medical Research, Vol 27, Issue 9, 2018 https://journals.sagepub.com/doi/10.1177/0962280216683570

4. StataCorp. 2021. *Stata Statistical Software: Release 17*. College Station, TX: StataCorp LLC.

16 Appendix

16.1 Derived Patient Reported Outcome Measures (PROMs)

The PROMs are p-POSSUM, OBAS, QoR-15, PONV and EQ-5D. Codes developed in-house will be checked and validated by an independent statistician using dummy data. Table 16.1 describes how each score will be calculated.

Table 16.1 Calculation of PROMs scores

PROMs	Calculation
p-POSSUM	p-POSSUM will be calculated as follows which has been confirmed by the authors of the published paper[1]: • A numerical score of 1, 2, 4 and 8 is assigned to each of the physiological (12 variables) and operative severity variables (6 variables). • The total score is calculated by adding up the score for each question for physiological and operative severity parts separately.

	• Once these scores are known, the predicted risk for mortality will be estimated using the following equation (where R1 indicates mortality): $\text{Log}_e R1/(1\text{-}R1) = -7.04 + (0.13 \times \text{Physiological Score}) + (0.16 \times \text{Operative Severity Score})$ • If any question is not answered (i.e. there is missing data) then the score cannot be calculated.
OBAS	To calculate the OBAS score, compute the sum of scores in items 1–6 and add ‘4−score in item 7’. Note that a low score indicates high benefit. If any question is not answered (i.e. there is missing data) then the score cannot be calculated[2].
Quality of Recovery - 15 (QoR-15)	To calculate the QoR-15 score, compute the sum of scores. The score ranges from 0 to 150 with a higher score indicating a better QoR. If any question is not answered (i.e. there is missing data) then the score cannot be calculated.[3]
PONV	To calculate the PONV Impact Scale score, add the numerical responses to questions 1 and 2. A PONV Impact Scale score of =5 defines clinically important PONV[4].
EQ-5D	The validated Stata code from EuroQol crosswalk to the 3L tool maybe found for United Kingdom here: https://euroqol.org/support/analysis-tools/cross-walk/. This is required currently by NICE.

16.2 Derived Composite Outcome Measures

The composite measures are GI-2 and GI-3. Codes developed in-house will be checked and validated by an independent statistician using dummy data. Table 16.2 describes how each score will be calculated.

Table 16.2 Calculation of composite outcome scores

Composite outcomes	Calculation
G1-2 recovery	GI-2 recovery a composite endpoint defined as the achievement of both of the following two events: 1. tolerating diet without significant nausea or vomiting for 3 consecutive meals AND 2. first passage of stool Participant will be coded as has achieved or not achieved GI-2 recovery by 96 hours after the start of operation
G1-3 recovery	GI-3 recovery is a composite endpoint defined as the achievement of both of the following two events: 1. tolerating diet without significant nausea or vomiting for 3 consecutive meals AND 2. passage of flatus OR stool Participant will be coded as has achieved or not achieved GI-3 recovery by 72 hours after the start of operation

Reference:

1. Copeland GP, Jones D, Walters M. POSSUM: a scoring system for surgical audit. Br J Surg. 1991 Mar;78(3):355-60. doi: 10.1002/bjs.1800780327. PMID: 2021856.
2. Lehmann N, Joshi GP, Dirkmann D, Weiss M, Gulur P, Peters J, Eikermann M. Development and longitudinal validation of the overall benefit of analgesia score: a simple multi-dimensional quality assessment instrument. Br J Anaesth. 2010 Oct;105(4):511-8. doi: 10.1093/bja/aeq186. Epub 2010 Aug 6. PMID: 20693179.
3. Stark PA, Myles PS, Burke JA. Development and psychometric evaluation of a postoperative quality of recovery score: the QoR-15. Anesthesiology. 2013 Jun;118(6):1332-40. doi: 10.1097/ALN.0b013e318289b84b. PMID: 23411725.
4. Myles PS, Wengritzky R. Simplified postoperative nausea and vomiting impact scale for audit and post-discharge review. Br J Anaesth. 2012 Mar;108(3):423-9. doi: 10.1093/bja/aer505. Epub 2012 Jan 29. PMID: 22290456.

16.3 Stata code

Table 16.2 provides sample Stata code for the analysis of each outcome

Table 16.2: Stata code

<table>
<tr><td>Outcome</td><td>Stata code</td></tr>
<tr><td>Primary outcome:</td><td></td></tr>
<tr><td>GI-3 recovery</td><td><code>meglm GI3 i.IV i.AgeBand i.Male || CentreNo:, family(poisson) link(log) vce(robust) eform</code>
<code>GI3</code> binary (0 for not achieving GI3 and 1 for achieving GI3)
<code>IV</code> binary (0 for placebo and 1 for IV lidocaine)
<code>AgeBand</code> categorical (<50 years, 50-74 years, 75 years and older)
<code>Male</code> binary (0 for female and 1 for male)
<code>CentreNo</code> categorical (corresponds to each recruiting centre)</td></tr>
<tr><td>Secondary outcomes:</td><td></td></tr>
<tr><td>Time to achieving GI-2</td><td><code>stset Time, failure(GI2)</code>

<code>stcox i.IV i.AgeBand i.Male, shared(CentreNo)</code>
<code>Time</code> continuous (From time of start of general anaesthetic to time of achieving GI2)
<code>GI2</code> binary (0 for not achieving GI2 and 1 for achieving GI2)
<code>IV</code> binary (0 for placebo and 1 for IV lidocaine)
<code>AgeBand</code> categorical (<50 years, 50-74 years, 75 years and older)
<code>Male</code> binary (0 for female and 1 for male)
<code>CentreNo</code> categorical (corresponds to each recruiting centre)</td></tr>
<tr><td>Time to achieving GI-3</td><td><code>stset Time, failure(GI3)</code>

<code>stcox i.IV i.AgeBand i.Male, shared(CentreNo)</code>
<code>Time</code> continuous (From time of start of general anaesthetic to time of achieving GI3)
<code>GI2</code> binary (0 for not achieving GI3 and 1 for achieving GI3)
<code>IV</code> binary (0 for placebo and 1 for IV lidocaine)
<code>AgeBand</code> categorical (<50 years, 50-74 years, 75 years and older)
<code>Male</code> binary (0 for female and 1 for male)
<code>CentreNo</code> categorical (corresponds to each recruiting centre)</td></tr>
</table>

Prolonged post-operative Ileus - proportion of patients who failed to establish GI-3 by 120 hours after surgery	`meglm i.PPOI i.IV i.AgeBand i.Male || CentreNo:, family(poisson) link(log) vce(robust) eform` • `PPOI` binary (0 for established GI-3 by 120 hours and 1 for failed to establish GI3 by 120 hours - PPOI) • `IV` binary (0 for placebo and 1 for IV lidocaine) • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `CentreNo` categorical (corresponds to each recruiting centre)
Quality of analgesia	`memixed OBAS OBAS_baseline i.AgeBand i.Male i.IV##i.TimePoint || CentreNo: || StudyNo:, vce(robust)` • `OBAS` continuous • `OBAS` continuous baseline score • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `IV` binary (0 for placebo and 1 for IV lidocaine) • `TimePoint` categorical (coded 1 = Postoperative Day 1, 2 = Postoperative Day 2, 3 = Postoperative Day 3, 4 = Postoperative Day 4, 5 = Postoperative Day 5, 6 = Postoperative Day 6, 7 = Postoperative Day 7) • `CentreNo` categorical (corresponds to each recruiting centre) • `StudyNo` unique participant identifier.
Quality of recovery	`memixed QoR QoR_baseline i.AgeBand i.Male i.IV##i.TimePoint || CentreNo: || StudyNo:, vce(robust)` • `QoR` continuous • `QoR` continuous baseline score • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `IV` binary (0 for placebo and 1 for IV lidocaine) • `TimePoint` categorical (coded 1 = Postoperative Day 1, 2 = Postoperative Day 2, 3 = Postoperative Day 3, 4 = Postoperative Day 4, 5 = Postoperative Day 5, 6 = Postoperative Day 6, 7 = Postoperative Day 7, 30 = Postoperative Day 30) • `CentreNo` categorical (corresponds to each recruiting centre) • `StudyNo` unique participant identifier.

EQ-5D	`memixed EQ5D EQ5D_baseline i.AgeBand i.Male i.IV##i.TimePoint || CentreNo: || StudyNo:, vce(robust)` • `EQ5D` continuous • `EQ5D` continuous baseline score • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `IV` binary (0 for placebo and 1 for IV lidocaine) • `TimePoint` categorical (coded 1 = Postoperative Day 1, 2 = Postoperative Day 2, 3 = Postoperative Day 3, 4 = Postoperative Day 4, 5 = Postoperative Day 5, 6 = Postoperative Day 6, 7 = Postoperative Day 7, 30 = Postoperative Day 30, 90 = Postoperative Day 90) • `CentreNo` categorical (corresponds to each recruiting centre) • `StudyNo` unique participant identifier.
Time to achieving medical criteria for discharge	`meglm Time_medicalcriteria i.IV i.AgeBand i.Male || CentreNo:, vce(robust)` • `Time_medicalcriteria` continuous (From time of start of general anaesthetic to time of discharge) • `IV` binary (0 for placebo and 1 for IV lidocaine) • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `CentreNo` categorical (corresponds to each recruiting centre)
Time to patient reported readiness for discharge	`meglm Time_readiness i.IV i.AgeBand i.Male || CentreNo:, vce(robust)` • `Time_readiness` continuous (From time of start of general anaesthetic to time of patient reported readiness for discharge) • `IV` binary (0 for placebo and 1 for IV lidocaine) • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `CentreNo` categorical (corresponds to each recruiting centre)
Tertiary outcomes:	
Total length of hospital stay	`meglm total_LOS i.IV i.AgeBand i.Male || CentreNo:, vce(robust)` • `total_LOS` continuous (number of days in hospital)

	• `IV` binary (0 for placebo and 1 for IV lidocaine) • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `CentreNo` categorical (corresponds to each recruiting centre)
Subgroup analysis	
6hr v 12 hours IMP infusion (defined by what was intended)	`meglm i.GI3 i.IV##i.IMP i.AgeBand i.Male ||CentreNo:, family(poisson) link(log) vce(robust) eform` • `GI3` binary (0 for not achieving GI3 and 1 for achieving GI3) • `IMP` binary (0 for 6hr and 1 for 12hr) • `IV` binary (0 for placebo and 1 for IV lidocaine) • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male)
By operation (right v non-right hemicolectomy)	`meglm i.GI3 i.IV##i.Op i.AgeBand i.Male || CentreNo:, family(poisson) link(log) vce(robust) eform` • `GI3` binary (0 for not achieving GI3 and 1 for achieving GI3) • `Op` binary (0 for right and 1 for non-right) • `IV` binary (0 for placebo and 1 for IV lidocaine) • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `CentreNo` categorical (corresponds to each recruiting centre)
Gender (M vs F)	`meglm i.GI3 i.IV##i.Male i.AgeBand || CentreNo:, family(poisson) link(log) vce(robust) eform` • `GI3` binary (0 for not achieving GI3 and 1 for achieving GI3) • `IV` binary (0 for placebo and 1 for IV lidocaine) • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `CentreNo` categorical (corresponds to each recruiting centre)

Age (<50 years, 50-74 years, 75 years and older)	`meglm i.GI3 i.IV##i.AgeBand i.Male || CentreNo:, family(poisson) link(log) vce(robust) eform` • `GI3` binary (0 for not achieving GI3 and 1 for achieving GI3) • `IV` binary (0 for placebo and 1 for IV lidocaine • `AgeBand` categorical (<50 years, 50-74 years, 75 years and older) • `Male` binary (0 for female and 1 for male) • `CentreNo` categorical (corresponds to each recruiting centre)
ERAS compliance (High, moderate, low)	`meglm i.GI3 i.IV##i.ERAS i.Male || CentreNo:, family(poisson) link(log) vce(robust) eform` • `GI3` binary (0 for not achieving GI3 and 1 for achieving GI3) • `IV` binary (0 for placebo and 1 for IV lidocaine • `ERAS` categorical (High, moderate, low) • `Male` binary (0 for female and 1 for male) • `CentreNo` categorical (corresponds to each recruiting centre)

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